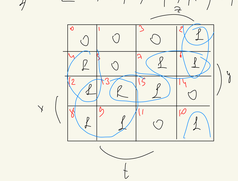
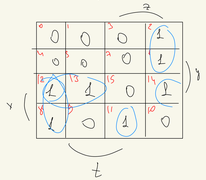
x	y	z	t	a	b	c	d	e	f	g	No	
0	0	0	0	0	0	0	0	0	0	0	0	OFF
0	0	0	1	0	0	1	0	1	0	1	1	n
0	0	1	0	1	1	1	0	1	1	1	2	a
0	0	1	1	0	0	0	0	1	0	1	3	z
0	1	0	0	1	0	1	1	1	1	0	4	g
0	1	0	1	0	0	1	0	0	0	0	5	i
0	1	1	0	1	1	0	1	1	0	1	6	z
0	1	1	1	1	0	1	0	1	0	1	7	m
1	0	0	0	1	1	1	0	1	1	1	8	a
1	0	0	1	1	0	1	0	1	0	1	9	m
1	0	1	0	1	0	0	1	1	1	1	10	e
1	0	1	1	0	1	1	1	1	0	1	11	d
1	1	0	0	1	1	0	1	1	0	1	12	z
1	1	0	1	1	1	1	0	1	1	1	13	a
1	1	1	0	0	1	1	1	1	0	1	14	d
1	1	1	1	1	0	0	1	1	1	1	15	e

1) $a = \Sigma\{2,4,6,7,8,9,10,12,13,15\}$



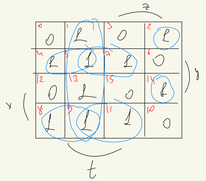
$SP = xz' + yz't' + xyt + x'yz + x'zt' + y'zt'$

2) $b = \Sigma\{2,6,8,11,12,13,14\}$



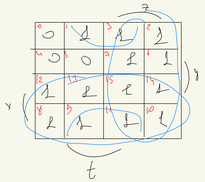
$xz't' + xyz' + xyt' + x'yzt + x't'z$

3) $c = \Sigma\{1,2,4,5,7,8,9,11,13,14\}$



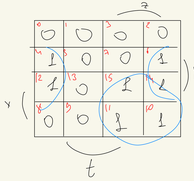
$xy'z' + xy't + z't + x'yz + x'yt + x'y'zt + xyzt'$

7) $g = \Sigma\{1,2,3,6,7,8,9,10,11,12,13,14,15\}$



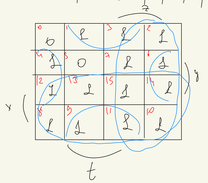
$x + z + y't$

4) $d = \Sigma\{4,6,10,11,12,14,15\}$



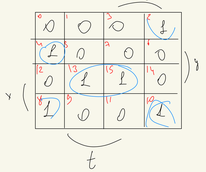
$yt' + xz$

5) $e = \Sigma\{1,2,3,4,6,7,8,9,10,11,12,13,14,15\}$



$x + z + yt' + y't$

6) $f = \Sigma\{2,4,8,10,13,15\}$



$xyt + xy't' + y'zt' + x'yz't'$

$a = xz' + yz't' + xyt + x'yz + x'zt' + y'zt' = t'(y \oplus z)$

$b = xz't' + xyz' + xyt' + xy'zt + x't'z = xy(t' + z')$

$c = xy'z' + xyt' + z't + x'yz' + x'yt + x'y'zt' + xyzt' = z'(x \oplus y) + t(x \oplus y) + zt'(x \odot y)$

$d = yt' + xz$

$e = x + z + yt' + y't = x + z + y \oplus t$

$f = xyt + xy't' + y'zt' + x'yz't' = x(y \odot t) + y'zt' + x'yz't'$

$g = x + z + y't$